

Lecture 1 Introduction to Multivariable Calculus

Instructor: Xuechao Wang FTEC/IOT

xuechao.wang@hkust-gz.edu.cn W1 415

OH: Every Tuesday 15:00-16:00, W1 415

TA: Qianyu Yu 3rd year FTEC PhD

qyu100@connect.hkust-gz.edu.cn

USTC → Fudan

Hongxu Su, 2nd year FTEC PhD

hsu238@connect.hkust-gz.edu.cn

ASU → NUS

Tutorial: Every Friday 13:30-14:20, E1 148, delivered by TAs

Grading: 30% weekly assignments + 30% midterm + 40% final

Homework: Roughly 12 weekly assignments, each containing 6-8 problems.

3pts each

Sum up the highest 10 attempts

Due every Thursday before class

No late submission or re-submission is allowed.

Final grades:

Absolute score	Letter grade
[85, 100]	A- or A or A+
[70, 85)	B- or B or B+
[55, 70)	C- or C or C+
[40, 55)	D
[0, 40)	F

Bonus questions with extra points may or may not be provided

Course materials: self-contained lecture notes will be provided

Recommended textbooks:

- 1) Calculus for Scientists and Engineers
by Briggs, Cochran, Gillett
- 2) Thomas's Calculus: Early Transcendentals 13th Edition.

Tentative schedule:

- Week 1: Lines, Planes and Curves in 3D } topics of the midterm
- Week 2-5: Partial Differentiations
- Week 6-8: Multiple Integrations
- Week 9-13: Vector Calculus.

Recap on Calculus.

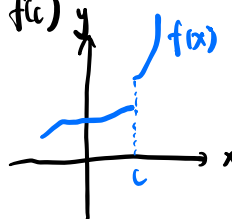
$f(x)$, $f'(x)$, $\int_a^b f(x) dx$
Continuous function Derivative Riemann Integral.

f is continuous at some point c if $\lim_{x \rightarrow c} f(x) = f(c)$

This means ① f is defined at c

② the limit of that equation exists, i.e. $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x)$

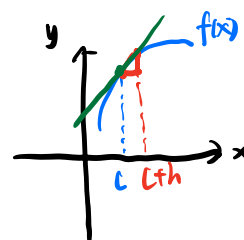
③ the value of this limit equals $f(c)$



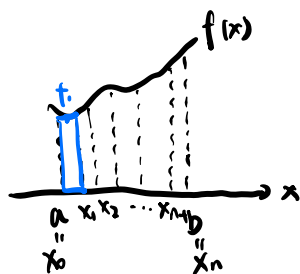
The derivative of f at some point c is

$$f'_x|_{x=c} = \left. \frac{df}{dx} \right|_{x=c} = f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

If this limit exists, then f is called differentiable at point c



Riemann integral:



$$a = x_0 \leq t_1 \leq x_1 \leq t_2 \leq x_2 \leq \dots \leq x_{n-1} \leq t_n \leq x_n = b$$

$$\Delta_i = x_i - x_{i-1} \quad \text{mesh} \quad \Delta = \max_{1 \leq i \leq n} \Delta_i$$

$$\int_a^b f(x) dx = \lim_{\Delta \rightarrow 0} \sum_{i=1}^n f(t_i) \Delta_i$$

if this limit exists, then f is called Riemann integrable in $[a, b]$

Fundamental theorem of Calculus.

$$\int_a^b f'(t) dt = f(b) - f(a)$$

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Q. Why is single-variable Calculus not enough?

Example. Neural Network Training

= Multivariable Optimization.

$$f_\theta: x \rightarrow y$$

$$\theta = (w_1, w_2, \dots, w_n)$$

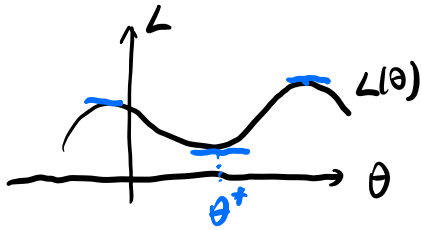
$$\text{Loss function: } L(\theta) = \frac{1}{m} \sum_{i=1}^m \ell(\underbrace{f_\theta(x^{(i)})}_{\text{output of NN}}, \underbrace{y^{(i)}}_{\text{true label}})$$

data set $(x^{(i)}, y^{(i)})_{i=1, \dots, m}$

ℓ = loss for a single data point, e.g. $\ell(\hat{y}, y) = (\hat{y} - y)^2$

$$\text{Goal of NN training: } \min_{\theta} L(\theta)$$

If θ only has one parameter, i.e., $\theta \in \mathbb{R}$, then this is easy



But in practice θ has billions of parameters, so we need multivariable Calculus

$$\theta \in \mathbb{R}^n$$

Gradient $\nabla_{\theta} L(\theta) = \left(\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \dots, \frac{\partial L}{\partial w_n} \right)$

Gradient Descent. $\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta)$